NON-FUNCTIONAL REQUIREMENTS

21 measurable requirements (NF-01 to NF-21), each with its own acceptance test.

20 s

Performance

NF-01 – NF-02

- Node state change on the map in ≤ 20 s
- 95 % of history queries ≤ 2 s with 50 concurrent users

R/O

Security

NF-03 – NF-06

- Colbits telemetry and alarm tables are read-only
- Salted bcrypt / Argon2 hashes
- Lock 15 min after 5 failed logins
- Token expires after 15 idle minutes

6

Flexibility

NF-07 – NF-12

- Configurable with no code change:
- inactivity and low-energy thresholds
- 5-min grouping window, min. duration
- severity criterion, SMAC tolerance

80 %

Testability

NF-13 – NF-21

- API + UI suites on generated data
- Explicit scope-leak test
- CI on every PR to main
- ≥ 80 % API coverage, clean-env setup

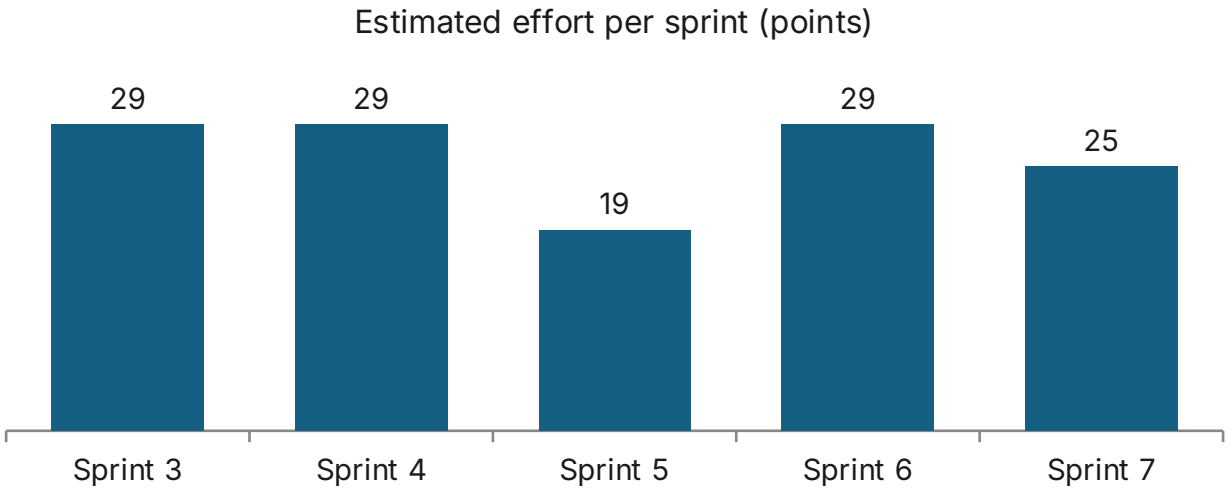
INITIAL PRODUCT BACKLOG

29 user stories, all To Do — coding starts in Sprint 3.

29 user stories, each traced to F- / NF- requirements

131 estimated effort points

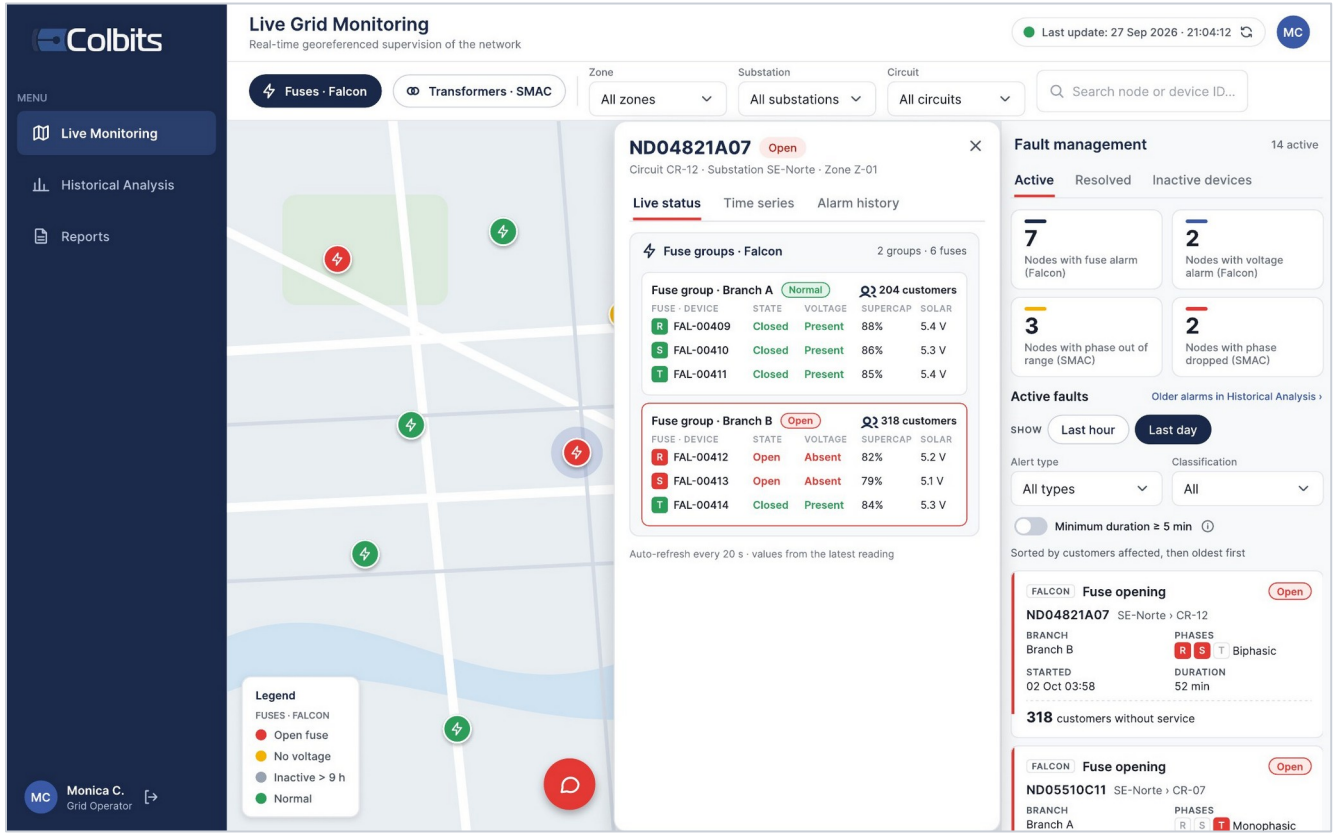
5 sprints planned (Sprint 3 to 7)



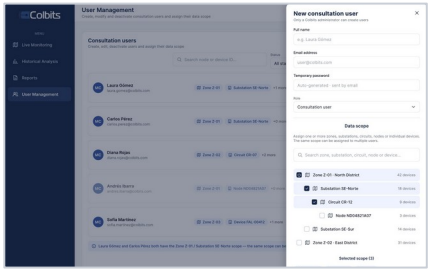
Priority 5	**US1–US12** Generator, login, scope, map, inactivity, fuse and transformer colors · Sprints 3–4
Priority 4	**US13–US26** Filters, faults and alarms, user management, history, indicators, export · Sprints 5–7
Priority 3–2	**US27–US29** Alarm → map, momentary openings, Spanish chat (optional) · Sprint 7

INITIAL MOCKUPS (FIGMA)

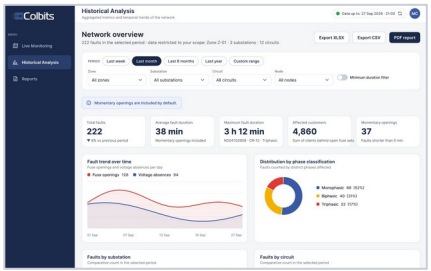
COLBITS Mockups v3 · built from Colbits' reference screens and reviewed with the client.



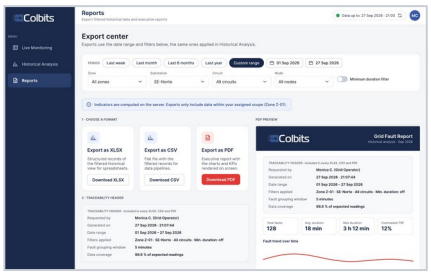
Live grid monitoring fuse / transformer layers · node detail · faults by severity



User management
The admin creates consultation users and assigns their data scope

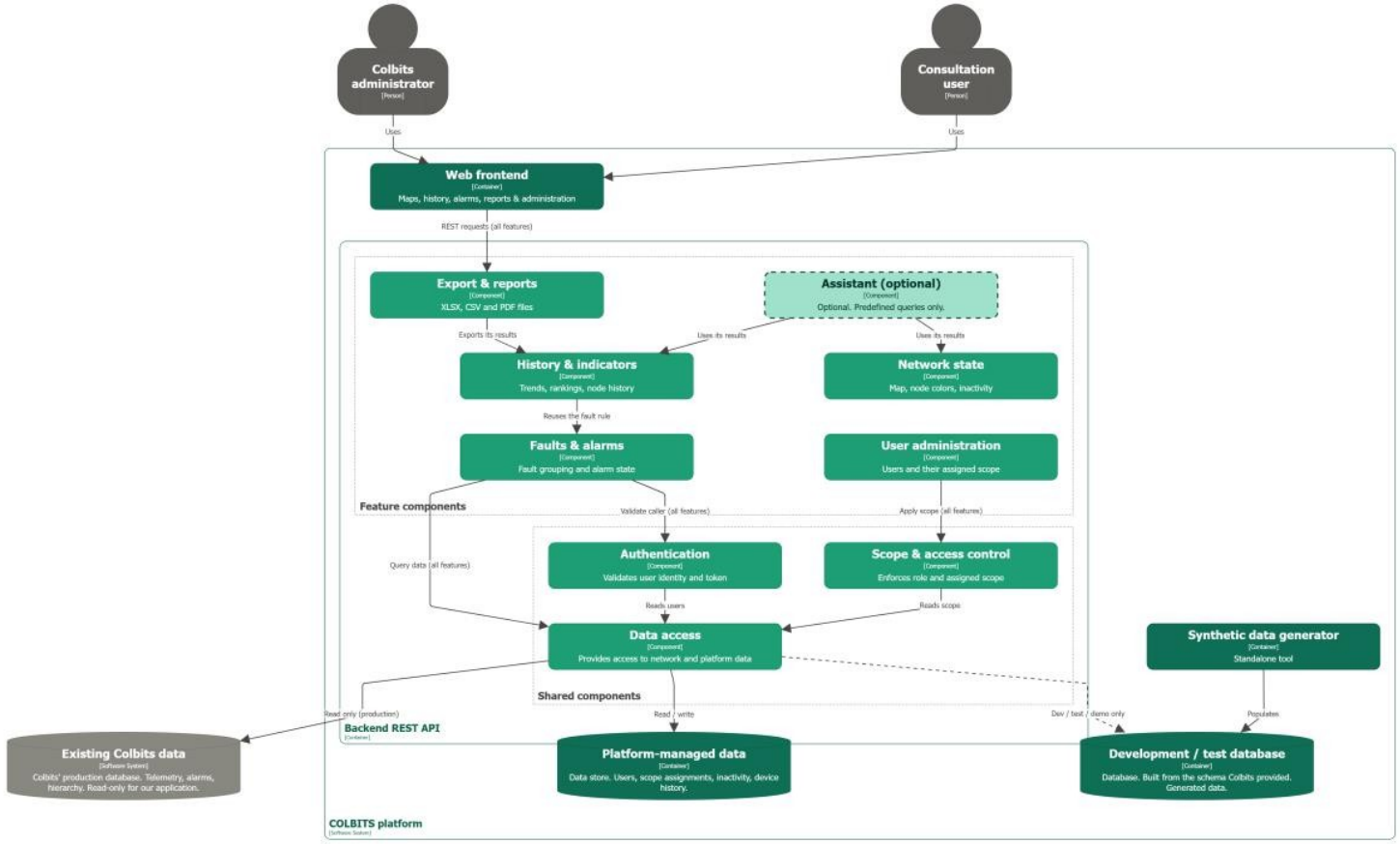


Historical analysis
Date presets, filters, KPIs, trends and rankings



Export center
XLSX · CSV · PDF with the traceability header

SYSTEM DECOMPOSITION



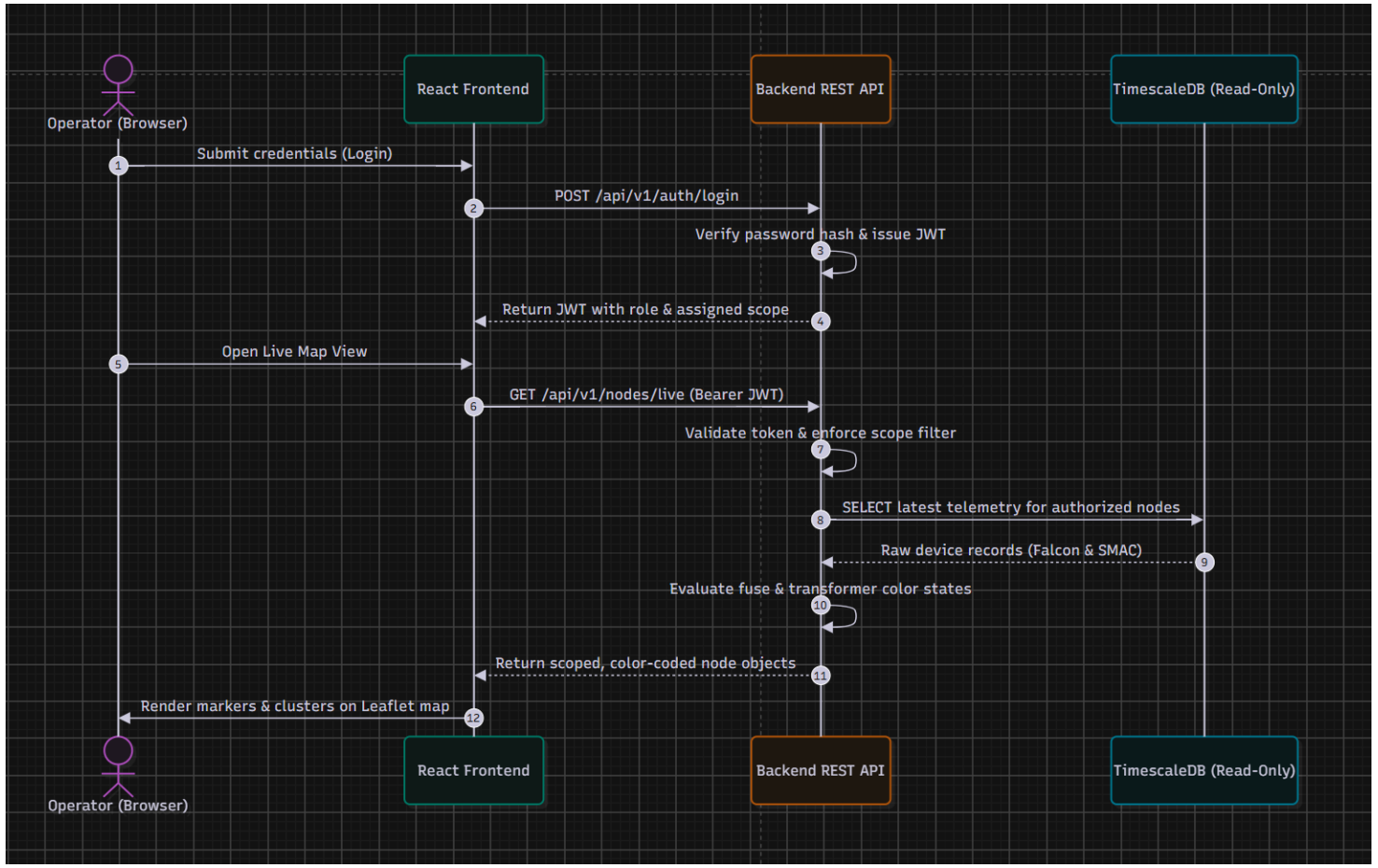
Legend

- Colbits already has it
- We build it
- Optional (dashed)

Key points

- Both roles use only the web frontend
- 6 feature components + 3 shared ones
- Colbits data: read only
- Our data: users, scope, faults, device history
- Generator fills a separate dev / test DB

SEQUENCE 1 · SECURE LIVE MAP



1. **Login**
   password hash checked; JWT with role and scope

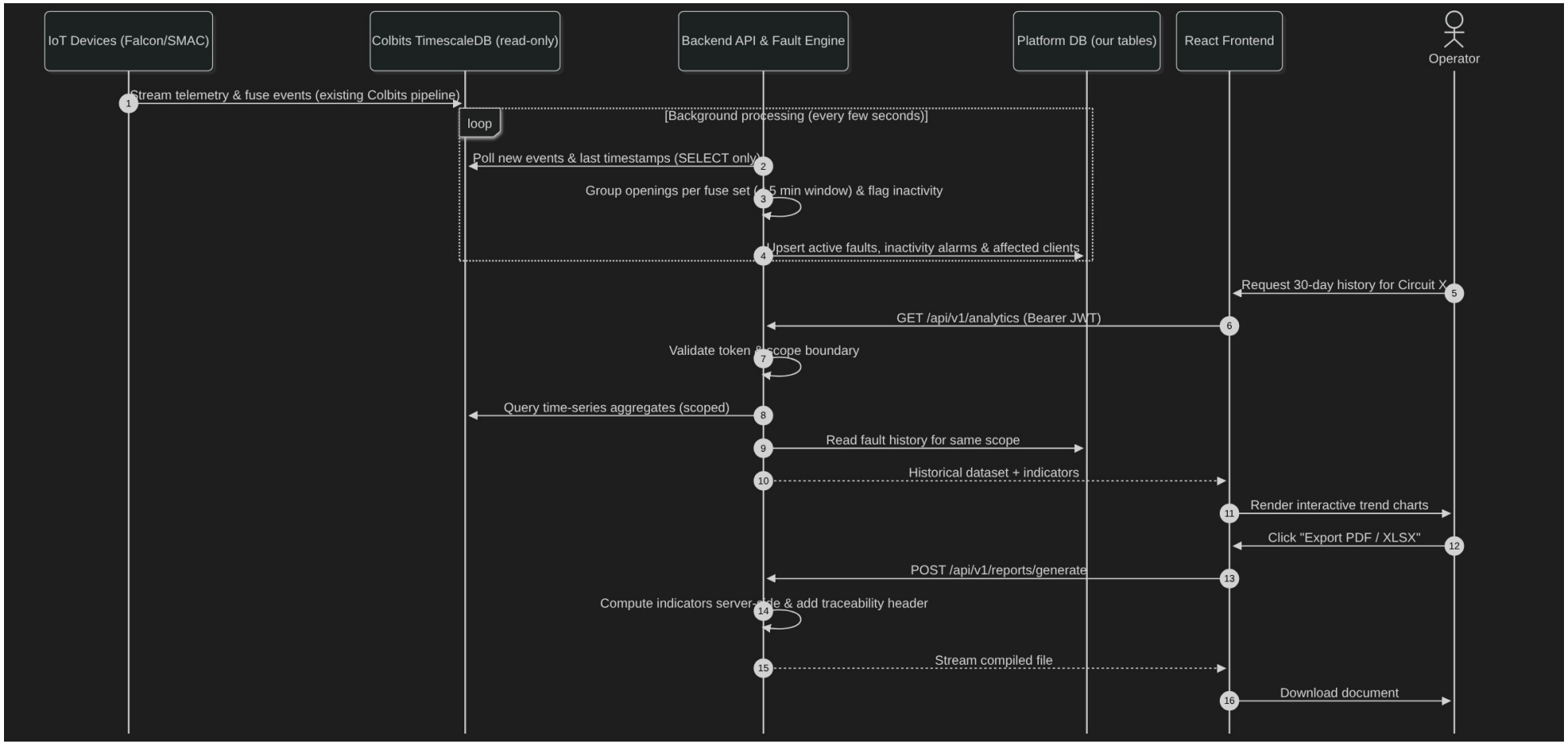
2. **Scoped request**
   every call carries the token

3. **Scope filter**
   applied on the server, before any query

4. **Read-only query**
   latest Falcon & SMAC telemetry

5. **Node colors**
   computed in the API, drawn on the map

SEQUENCE 2 · FAULTS & REPORTS



KEY ARCHITECTURAL DECISIONS

Decision	Why	Rejected alternative
Modular monolith: one REST API with feature modules	6 part-time devs, 14 weeks: one deploy, one test suite; modules mirror the diagram	Microservices
Colbits data read-only; our state in a separate schema	Production stays untouched (NF-03); enforced by a SELECT-only DB role	New tables or columns in Colbits' schema
Scope enforced once, on the server, in the data-access layer	No leak through counts or totals (RF-SEG-08); one place to test (NF-14)	Filtering per endpoint or in the frontend
One fault rule, run by a background engine	Same numbers on map, list, history, export and chat; fast queries (NF-02)	Recomputing faults on every request
Live map by short polling (≤ 10 s)	Meets the 20 s target simply; easy "last refresh" warning (RF-MAP-11)	WebSockets / SSE (revisit if needed)
Seeded generator + Docker + CI from day one	Reproducible dev and tests without real data (NF-16, NF-18)	Testing on the shared DB copy

OPEN TECHNICAL CONCERNS

Performance on a database we cannot change

No new indexes or aggregates in Colbits' DB → pre-aggregate in our schema; load-test NF-02 early

Real-data quirks

SMAC readings split over 3 tables (time-tolerance join); legacy repetition fuse IDs the generator does not produce

End-to-end 20 s latency

Poll interval + engine cycle + DB time must stay under 20 s → measured with the NF-01 test

Open points with Colbits

Amber vs. yellow and the legend states in the mockups; DB copy with anonymised coordinates

TECHNOLOGIES BY COMPONENT

Web frontend

Web frontend
React + TypeScript (Vite), Atomic Design

Map
Leaflet + react-leaflet + markercluster

Charts
Recharts

Live refresh
TanStack Query, 10 s polling

Backend REST API

Feature components
Python 3.12 + FastAPI + Pydantic

Authentication
JWT + Argon2

Scope & data access
SQLAlchemy 2 + psycopg, SELECT-only role

Faults & alarms engine
APScheduler background job

Export & reports
openpyxl · csv · WeasyPrint

Data, tooling & quality

Colbits data · dev / test DB
PostgreSQL + TimescaleDB

Platform-managed data
PostgreSQL schema + Alembic

Synthetic data generator
Python CLI (Typer + seeded NumPy)

Tests & CI
pytest + coverage · Playwright · GitHub Actions

Packaging
Docker + Docker Compose